# Graph Theory & Networks (AI HL)

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**ANSWER KEY — COMPLETE WORKED SOLUTIONS**

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** **[Foundational]** • **Target: Degrees and the Handshaking Lemma**

A weighted graph has edges (with weights) $1–2(1), 2–3(6), 3–4(5), 4–5(7), 1–5(8), 2–5(2), 1–3(9)$. Find the total weight of the minimum spanning tree.

Final Answer: 14. Applying Kruskal's algorithm (adding edges in increasing weight order, skipping cycles) gives a minimum spanning tree of total weight 14.

**2** **[Foundational]** • **Target: Eulerian and Hamiltonian Paths and Circuits**

A weighted graph has edges (with weights) $1–2(9), 2–3(5), 3–4(7), 4–5(2), 1–5(3), 2–5(4), 1–3(8)$. Find the total weight of the minimum spanning tree.

Final Answer: 14. Applying Kruskal's algorithm (adding edges in increasing weight order, skipping cycles) gives a minimum spanning tree of total weight 14.

**3** **[Foundational]** • **Target: Minimum Spanning Trees (Kruskal & Prim)**

A weighted graph has edges (with weights) $1–2(8), 2–3(6), 3–4(3), 4–5(1), 1–5(4), 2–5(7), 1–3(9)$. Find the total weight of the minimum spanning tree.

Final Answer: 14. Applying Kruskal's algorithm (adding edges in increasing weight order, skipping cycles) gives a minimum spanning tree of total weight 14.

**4** **[Foundational]** • **Target: Shortest Paths and Dijkstra's Algorithm**

A weighted graph has edges (with weights) $1–2(3), 2–3(1), 3–4(2), 4–5(4), 1–5(5), 2–5(6), 1–3(9)$. Find the total weight of the minimum spanning tree.

Final Answer: 10. Applying Kruskal's algorithm (adding edges in increasing weight order, skipping cycles) gives a minimum spanning tree of total weight 10.

**5** **[Foundational]** • **Target: Degrees and the Handshaking Lemma**

A weighted graph has edges (with weights) $1–2(1), 2–3(6), 3–4(3), 4–5(2), 1–5(4), 2–5(5), 1–3(8)$. Find the total weight of the minimum spanning tree.

Final Answer: 10. Applying Kruskal's algorithm (adding edges in increasing weight order, skipping cycles) gives a minimum spanning tree of total weight 10.

**6** **[Foundational]** • **Target: Eulerian and Hamiltonian Paths and Circuits**

A weighted graph has edges (with weights) $1–2(1), 2–3(4), 3–4(3), 4–5(5), 1–5(6), 2–5(7), 1–3(8)$. Find the total weight of the minimum spanning tree.

Final Answer: 13. Applying Kruskal's algorithm (adding edges in increasing weight order, skipping cycles) gives a minimum spanning tree of total weight 13.

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**7** [Foundational] • Target: Minimum Spanning Trees (Kruskal & Prim)

A weighted graph has edges (with weights) $1{-}2(6), 2{-}3(8), 3{-}4(2), 4{-}5(4), 1{-}5(7), 2{-}5(9), 1{-}3(10)$. Find the total weight of the minimum spanning tree.

Final Answer: 19. Applying Kruskal's algorithm (adding edges in increasing weight order, skipping cycles) gives a minimum spanning tree of total weight 19.

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## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Shortest Paths and Dijkstra's Algorithm

Consider the function $f(x) = \frac{1}{2}x^2 - 3x + 5$. The graph of $y = f(x)$ is a parabola. Find the x-coordinate of the vertex of the parabola.

Final Answer: 3. The x-coordinate of the vertex of a parabola given by the equation $f(x) = ax^2 + bx + c$ is found using the formula $x = -\frac{b}{2a}$. In this case, $a = \frac{1}{2}$ and $b = -3$. Therefore, the x-coordinate of the vertex is $x = -\frac{-3}{2(\frac{1}{2})} = -\frac{-3}{1} = 3$.

**9** [Intermediate] • Target: Degrees and the Handshaking Lemma

State the principal value (in degrees, $-90^\circ \le \theta \le 90^\circ$) satisfying $\sin\theta = -\frac{1}{2}$.

Final Answer: $-30^\circ$. Working through this additional-topics step: $\sin^{-1}(-\frac{1}{2}) = -30^\circ$ (within the required range $-90^\circ \le \theta \le 90^\circ$; $150^\circ$ is out of range).

**10** [Intermediate] • Target: Eulerian and Hamiltonian Paths and Circuits

If $z = 2(\cos 45^\circ + i\sin 45^\circ)$, find $|z^6|$ and $\arg(z^6)$ (in degrees, $0 \le \arg < 360$).

Final Answer: $|z^6| = 64,\ \arg(z^6) = 270^\circ$. By De Moivre's theorem, $|z^6| = |z|^6 = 2^6 = 64,\ \arg(z^6) = 6 \times 45^\circ = 270^\circ$

**11** [Intermediate] • Target: Minimum Spanning Trees (Kruskal & Prim)

If $z = 2(\cos 30^\circ + i\sin 30^\circ)$, find $|z^6|$ and $\arg(z^6)$ (in degrees, $0 \le \arg < 360$).

Final Answer: $|z^6| = 64,\ \arg(z^6) = 180^\circ$. By De Moivre's theorem, $|z^6| = |z|^6 = 2^6 = 64,\ \arg(z^6) = 6 \times 30^\circ = 180^\circ$

**12** [Intermediate] • Target: Shortest Paths and Dijkstra's Algorithm

If $z = 1(\cos 60^\circ + i\sin 60^\circ)$, find $|z^3|$ and $\arg(z^3)$ (in degrees, $0 \le \arg < 360$).

Final Answer: $|z^3| = 1,\ \arg(z^3) = 180^\circ$. By De Moivre's theorem, $|z^3| = |z|^3 = 1^3 = 1,\ \arg(z^3) = 3 \times 60^\circ = 180^\circ$

**13** [Intermediate] • Target: Degrees and the Handshaking Lemma

If $z = 1(\cos 90^\circ + i\sin 90^\circ)$, find $|z^6|$ and $\arg(z^6)$ (in degrees, $0 \le \arg < 360$).

Final Answer: $|z^6| = 1,\ \arg(z^6) = 180^\circ$. By De Moivre's theorem, $|z^6| = |z|^6 = 1^6 = 1,\ \arg(z^6) = 6 \times 90^\circ = 180^\circ$

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**14** [Intermediate] • Target: Eulerian and Hamiltonian Paths and Circuits

If $z = 2(\cos 60^\circ + i\sin 60^\circ)$, find $|z^3|$ and $\arg(z^3)$ (in degrees, $0 \le \arg < 360$).

Final Answer: $|z^3| = 8,\ \arg(z^3) = 180^\circ$. By De Moivre's theorem, $|z^3| = |z|^3 = 2^3 = 8,\ \arg(z^3) = 3 \times 60^\circ = 180^\circ$

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## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** [Challenging] • Target: Minimum Spanning Trees (Kruskal & Prim) • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

If $z = 1(\cos 45^\circ + i\sin 45^\circ)$, find $|z^5|$ and $\arg(z^5)$ (in degrees, $0 \le \arg < 360$).

Final Answer: $|z^5| = 1,\ \arg(z^5) = 225^\circ$. By De Moivre's theorem, $|z^5| = |z|^5 = 1^5 = 1,\ \arg(z^5) = 5 \times 45^\circ = 225^\circ$

**16** [Challenging] • Target: Shortest Paths and Dijkstra's Algorithm • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

If $z = 2(\cos 60^\circ + i\sin 60^\circ)$, find $|z^6|$ and $\arg(z^6)$ (in degrees, $0 \le \arg < 360$).

Final Answer: $|z^6| = 64,\ \arg(z^6) = 0^\circ$. By De Moivre's theorem, $|z^6| = |z|^6 = 2^6 = 64,\ \arg(z^6) = 6 \times 60^\circ = 0^\circ$

**17** [Challenging] • Target: Degrees and the Handshaking Lemma • Cross-Lesson: 6.01 Matrices (AI HL)

If $z = 2(\cos 30^\circ + i\sin 30^\circ)$, find $|z^3|$ and $\arg(z^3)$ (in degrees, $0 \le \arg < 360$).

Final Answer: $|z^3| = 8,\ \arg(z^3) = 90^\circ$. By De Moivre's theorem, $|z^3| = |z|^3 = 2^3 = 8,\ \arg(z^3) = 3 \times 30^\circ = 90^\circ$

**18** [Challenging] • Target: Eulerian and Hamiltonian Paths and Circuits • Cross-Lesson: 6.03 Markov Chains (AI HL)

If $z = 1(\cos 60^\circ + i\sin 60^\circ)$, find $|z^6|$ and $\arg(z^6)$ (in degrees, $0 \le \arg < 360$).

Final Answer: $|z^6| = 1,\ \arg(z^6) = 0^\circ$. By De Moivre's theorem, $|z^6| = |z|^6 = 1^6 = 1,\ \arg(z^6) = 6 \times 60^\circ = 0^\circ$

**19** [Challenging] • Target: Minimum Spanning Trees (Kruskal & Prim) • Cross-Lesson: 1.01 Proof by Mathematical Induction

A graph has vertices $\{1, \ldots, 6\}$ and edges $1{-}5, 1{-}4, 4{-}5, 2{-}6, 2{-}5, 1{-}3$. Find the degree of vertex $2$.

Vertex 2 appears in 2 edge(s), so its degree is 2.

**20** [Challenging] • Target: Shortest Paths and Dijkstra's Algorithm • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

A graph has vertices $\{1, \ldots, 6\}$ and edges $1{-}2, 5{-}6, 2{-}5, 1{-}3, 3{-}5$. Find the degree of vertex $2$.

Vertex 2 appears in 2 edge(s), so its degree is 2.

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## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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**21** [Foundational] • Target: Degrees and the Handshaking Lemma

What does the Handshaking Lemma state for any undirected graph with $V$ vertices and $E$ edges?

**a. The sum of the degrees of all vertices equals twice the number of edges: $\sum \deg(v) = 2E$ ✓**

**b.** The sum of the degrees equals $E$

**c.** The number of vertices with odd degree is always odd

**d.** Every vertex must have degree at least 2

Because every edge connects two endpoints, each edge contributes exactly 2 to the sum of the degrees of the vertices: $\sum \deg(v) = 2E$.

**22** [Foundational] • Target: Degrees and the Handshaking Lemma

A graph has $6$ vertices, each having degree $3$. How many edges does the graph have?

**a.** $9$ ✓

**b.** $18$

**c.** $6$

**d.** $12$

$\sum \deg(v) = 6 \times 3 = 18$. By the Handshaking Lemma, $2E = 18 \Rightarrow E = 9$.

**23** [Foundational] • Target: Degrees and the Handshaking Lemma

Why can a graph never have an odd number of vertices of odd degree?

**a. Because $\sum \deg(v) = 2E$ is always even, the sum of odd degrees must be even, forcing an even count of odd vertices ✓**

**b.** Because edges cannot cross each other

**c.** By Euler's formula $V - E + F = 2$

**d.** Because every vertex must have at least one adjacent edge

The total sum of degrees is even ($2E$). The sum of even degrees is even, so the sum of the odd degrees must also be even, which requires an even number of odd terms.

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**24** [Foundational] • Target: Eulerian and Hamiltonian Paths and Circuits

What condition is both necessary and sufficient for a connected graph to contain an Eulerian circuit (a closed trail visiting every edge exactly once)?

**a. Every vertex has an even degree ✓**

**b.** Exactly two vertices have odd degree

**c.** The graph contains no cycles

**d.** The graph is a complete bipartite graph

Euler proved that a connected graph has an Eulerian circuit if and only if all vertices have even degree.

**25** [Foundational] • Target: Eulerian and Hamiltonian Paths and Circuits

If a connected graph has exactly two vertices of odd degree, what type of Eulerian walk does it contain?

**a. An Eulerian trail (open path visiting every edge once) that starts at one odd vertex and ends at the other ✓**

**b.** An Eulerian circuit returning to the start

**c.** No Eulerian trail or circuit can exist

**d.** A Hamiltonian cycle

Euler's theorem guarantees an open Eulerian trail if and only if exactly two vertices have odd degree; the trail must start at one odd vertex and terminate at the other.

**26** [Foundational] • Target: Eulerian and Hamiltonian Paths and Circuits

What is the difference between an Eulerian circuit and a Hamiltonian cycle?

**a. An Eulerian circuit visits every edge once; a Hamiltonian cycle visits every vertex once ✓**

**b.** An Eulerian circuit visits every vertex once; a Hamiltonian cycle visits every edge once

**c.** They are identical concepts in graph theory

**d.** Hamiltonian cycles apply only to directed trees

Eulerian paths/circuits pertain to visiting every edge, while Hamiltonian paths/cycles pertain to visiting every vertex.

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**27** [Foundational] • Target: Minimum Spanning Trees (Kruskal & Prim)

In a connected weighted graph with $V$ vertices, how many edges does any spanning tree contain?

**a.** $V - 1$ ✓

**b.** $V$

**c.** $V + 1$

**d.** $\frac{V(V-1)}{2}$

Any tree on $V$ vertices is minimally connected and contains no cycles, having precisely $V - 1$ edges.

**28** [Foundational] • Target: Minimum Spanning Trees (Kruskal & Prim)

How does Kruskal's algorithm select edges to construct a Minimum Spanning Tree (MST)?

**a.** **Sorts all edges by weight and greedily adds the smallest available edge that does not create a cycle** ✓

**b.** Grows a single tree by continuously adding the closest vertex to the current tree

**c.** Removes the largest cycle edges one by one until only a tree remains

**d.** Visits vertices in alphabetical order

Kruskal's algorithm is an edge-based greedy algorithm that adds edges in order of increasing weight, rejecting any edge that completes a cycle.

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Minimum Spanning Trees (Kruskal & Prim)

How does Prim's algorithm differ conceptually from Kruskal's algorithm?

**a.** **Prim grows a single connected tree from an arbitrary starting vertex, whereas Kruskal maintains a forest of components until unified** ✓

**b.** Prim works only on unweighted graphs

**c.** Kruskal finds the longest path rather than minimum tree

**d.** Prim can only be applied to planar graphs

Prim's algorithm builds a single growing connected tree vertex by vertex, whereas Kruskal's algorithm connects disjoint components across the whole graph.

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**30** [Intermediate] • Target: Shortest Paths and Dijkstra's Algorithm

What is the purpose of Dijkstra's algorithm?

**a. To find the shortest paths from a single source vertex to all other vertices in a network with non-negative edge weights ✓**

**b.** To find a minimum spanning tree

**c.** To determine if a graph has an Eulerian circuit

**d.** To find the maximum flow through a network

Dijkstra's algorithm is the standard greedy algorithm for computing single-source shortest paths on graphs with non-negative edge weights.

**31** [Intermediate] • Target: Shortest Paths and Dijkstra's Algorithm

Why does Dijkstra's algorithm fail if a network contains negative edge weights?

**a. A greedy choice once finalized could be improved later by a negative edge, invalidating the optimality assumption ✓**

**b.** Distances cannot be negative by definition

**c.** The algorithm enters an infinite cycle immediately

**d.** Square roots of negative weights cannot be computed

Dijkstra's greedy assumption assumes that once a vertex distance is finalized, no shorter path to it exists; negative edges violate this monotonic property.

**32** [Intermediate] • Target: Shortest Paths and Dijkstra's Algorithm

In a planar graph, if $V = 6$ and $E = 9$, how many faces $F$ does the graph enclose according to Euler's formula $V - E + F = 2$?

**a.** 5 ✓

**b.** 3

**c.** 4

**d.** 6

By Euler's planar formula $V - E + F = 2$: $6 - 9 + F = 2 \Rightarrow -3 + F = 2 \Rightarrow F = 5$.

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**33** [Intermediate] • Target: Degrees and the Handshaking Lemma

A graph has vertices $\{1, \ldots, 6\}$ and edges $1{-}5, 2{-}3, 5{-}6, 3{-}6, 1{-}6, 2{-}5$. Find the degree of vertex $5$.

**a.** **3** ✓ **b.** 5

**c.** 1 **d.** 4

Vertex 5 appears in 3 edge(s), so its degree is 3.

**34** [Intermediate] • Target: Eulerian and Hamiltonian Paths and Circuits

A graph has vertices $\{1, \ldots, 6\}$ and edges $2{-}4, 4{-}6, 2{-}6, 2{-}5, 1{-}3, 3{-}5$. Find the degree of vertex $4$.

**a.** **2** ✓ **b.** 4

**c.** 0 **d.** 1

Vertex 4 appears in 2 edge(s), so its degree is 2.

**35** [Intermediate] • Target: Minimum Spanning Trees (Kruskal & Prim)

A graph has vertices $\{1, \ldots, 6\}$ and edges $2{-}4, 1{-}2, 1{-}4, 5{-}6, 3{-}6, 2{-}5$. Find the degree of vertex $4$.

**a.** 4 **b.** 1

**c.** 0 **d.** **2** ✓

Vertex 4 appears in 2 edge(s), so its degree is 2.

**36** [Intermediate] • Target: Shortest Paths and Dijkstra's Algorithm

A graph has vertices $\{1, \ldots, 5\}$ and edges $1{-}5, 1{-}4, 4{-}5, 2{-}5, 3{-}5$. Find the degree of vertex $4$.

**a.** **2** ✓ **b.** 0

**c.** 1 **d.** 4

Vertex 4 appears in 2 edge(s), so its degree is 2.

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**37** [Intermediate] • Target: Degrees and the Handshaking Lemma

A graph has vertices $\{1, \ldots, 6\}$ and edges $2{-}4, 1{-}2, 4{-}5, 3{-}6, 1{-}6, 1{-}3$. Find the degree of vertex $6$.

**a.** **2** ✓ **b.** 1

**c.** 4 **d.** 0

Vertex 6 appears in 2 edge(s), so its degree is 2.

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Eulerian and Hamiltonian Paths and Circuits • Cross-Lesson: 6.03 Markov Chains (AI HL)

A graph has vertices $\{1, \ldots, 5\}$ and edges $2{-}4, 1{-}2, 3{-}4, 1{-}4, 2{-}5$. Find the degree of vertex $2$.

**a.** **3** ✓ **b.** 5

**c.** 4 **d.** 2

Vertex 2 appears in 3 edge(s), so its degree is 3.

**39** [Challenging] • Target: Minimum Spanning Trees (Kruskal & Prim) • Cross-Lesson: 1.01 Proof by Mathematical Induction

State the principal value (in degrees, $-90^\circ \le \theta \le 90^\circ$) satisfying $\sin\theta = \frac{1}{2}$.

**a.** $50^\circ$ **b.** $30^\circ$ ✓

**c.** $69^\circ$ **d.** $62^\circ$

Working through this additional-topics step: $\sin^{-1}\left(\frac{1}{2}\right) = 30^\circ$

**40** [Challenging] • Target: Shortest Paths and Dijkstra's Algorithm • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

Evaluate $\sin^{-1}\left(-\frac{1}{2}\right)$ in degrees.

**a.** $-30^\circ$ ✓ **b.** $-24^\circ$

**c.** $-32^\circ$ **d.** $-17^\circ$

Working through this functions step: $\sin^{-1}\left(-\frac{1}{2}\right) = -30^\circ$

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**41** [Challenging] • Target: Degrees and the Handshaking Lemma • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

Evaluate $\cos^{-1}\left(-\frac{1}{2}\right)$ in degrees.

**a.** $120^\circ$ ✓

**b.** $97^\circ$

**c.** $142^\circ$

**d.** $143^\circ$

Working through this functions step: $\cos^{-1}\left(-\frac{1}{2}\right) = 120^\circ$

**42** [Challenging] • Target: Eulerian and Hamiltonian Paths and Circuits • Cross-Lesson: 1.04 Partial Fractions

Evaluate $\sin^{-1}\left(\frac{1}{2}\right)$ in degrees.

**a.** $41^\circ$

**b.** $30^\circ$ ✓

**c.** $15^\circ$

**d.** $24^\circ$

Working through this functions step: $\sin^{-1}\left(\frac{1}{2}\right) = 30^\circ$

**43** [Challenging] • Target: Minimum Spanning Trees (Kruskal & Prim) • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

State the principal value (in degrees, $-90^\circ \le \theta \le 90^\circ$) satisfying $\sin\theta = 1$.

**a.** $59^\circ$

**b.** $90^\circ$ ✓

**c.** $89^\circ$

**d.** $82^\circ$

Working through this additional-topics step: $\sin^{-1}(1) = 90^\circ$

**44** [Challenging] • Target: Shortest Paths and Dijkstra's Algorithm • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

State the principal value (in degrees, $-90^\circ \le \theta \le 90^\circ$) satisfying $\sin\theta = 0$.

**a.** $16^\circ$

**b.** $0^\circ$ ✓

**c.** $-34^\circ$

**d.** $18^\circ$

Working through this additional-topics step: $\sin^{-1}(0) = 0^\circ$

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**45** [Challenging] • Target: Degrees and the Handshaking Lemma • Cross-Lesson: 3.01 3D Vectors & the Cross Product

Evaluate $\tan^{-1}(-1)$ in degrees.

**a.** $-55^\circ$

**b.** $-58^\circ$

**c.** $-35^\circ$

**d.** $-45^\circ$ ✓

Working through this functions step: $\tan^{-1}(-1) = -45^\circ$

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